

Question Number	Answer	Additional Guidance	Mark
1(a)	A description that contains five of the following: • pondweed set up with suitable apparatus to measure volume of gas (1) • use of at least five appropriate (stated) temperatures (1) • time for pondweed to acclimatise • sodium hydrogen carbonate added (in excess) (1) • volume of gas collected in {same/stated} time (1) • identification and control of one variable (1) • method of calculating rate (1)	Accept name of apparatus eg photosynthometer/ diagram of apparatus to measure volume eg funnel +tube +(gas)syringe (connected together) Within range 5 - 50 °C Accept correct formula/bicarbonate/potassium hydrogen carbonate Accept minutes/hours/ per unit time Accept: pH – buffer mass or length of pondweed – measure light intensity – bulb at set distance – heatshield – wavelength with lamps/filter Accept number of bubbles divided by time	Exp (5)

Question Number	Answer	Additional Guidance	Mark
1(b)	An explanation that includes three of the following: • water (split to) form oxygen (1) • by photolysis (1) • electrons removed from water (1) • two oxygen (atoms) combine to form {one oxygen molecule/O₂} (1)	Ignore light dependent reaction unqualified Accept equation for MP1 and 3 $\text{H}_2\text{O} \rightarrow 2\text{H}^+ + 2\text{e}^- + \frac{1}{2} \text{O}_2$ to produce hydroxides/OH⁻ hydroxides react to form water and oxygen	Exp (3)